A

Industrial Oriented Mini Project

LOAN ELIGIBILITY PREDICTION FOR BANKS USING MACHINE LEARNING

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE (AI&DS)

submitted by

M. RAJESH (23K81A7296)

K. SREE AKHIL (23K81A7292)

CH. RAM MAHIDEEP VARMA (23K81A7279)

Under the Guidance of

MR. V. VISHNU VARDHAN

ASSISTANT PROFESSOR

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE (AI&DS)

St. MARTIN'S ENGINEERING COLLEGE

UGC Autonomous

Affiliated to JNTUH, Approved by AICTE Accredited by NBA & NAAC A+, ISO 9001-2008 Certified

Dhulapally, Secunderabad-500 100

www.smec.ac.in

March - 2026

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CERTIFICATE

This is to certify that the Industrial Oriented Mini Project entitled **"LOAN ELIGIBILITY PREDICTION FOR BANKS USING MACHINE LEARNING"** is being submitted by **M. RAJESH (23K81A7296)**, **K. SREE AKHIL (23K81A7292)** and **CH. RAM MAHIDEEP VARMA (23K81A7279)** in fulfilment of the requirement for the award of degree of **BACHELOR OF TECHNOLOGY in ARTIFICIAL INTELLIGENCE & DATA SCIENCE (AI & DS)** is recorded of Bonafide work carried out by them. The result embodied in this report have been verified and found satisfactory.

Signature of the Guide	Signature of the HOD
MR. V. VISHNU VARDHAN	**DR. N. KRISHNAIAH**
Assistant Professor	Head of the Department
Department of AI&DS	Department of AI&DS

Internal Examiner **External Examiner**

Place:
Date:

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DECLARATION

We, the students of **"Bachelor of Technology in Department of Artificial Intelligence & Data Science (AI & DS) "**, session: 2023 - 2027, **St. Martin's Engineering College, Dhulapally, Kompally, Secunderabad**, hereby declare that the work presented in this project work entitled **"LOAN ELIGIBILITY PREDICTION FOR BANKS USING MACHINE LEARNING"** is the outcome of our own Bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. This result embodied in this project report has not been submitted in any university for award of any degree.

M. RAJESH	**(23K81A7296)**
K. SREE AKHIL	**(23K81A7292)**
CH. RAM MAHIDEEP VARMA	**(23K81A7279)**

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance have crowded our efforts with success.

First and foremost, we would like to express our deep sense of gratitude and indebtedness to our College Management for their kind support and permission to use the facilities available in the Institute.

We especially would like to express our deep sense of gratitude and indebtedness to **Prof. Dr. K. RAVINDRA**, Group Director, St. Martin's Engineering College, Dhulapally, Secunderabad, for permitting us to undertake this project.

We wish to record our profound gratitude to **Dr. M. SREENIVAS RAO**, Principal, St. Martin's Engineering College, for his motivation and encouragement.

We are also thankful to **Dr. N.KRISHNAIAH,** Head of the Department, Artificial Intelligence and Data Science (AI & DS), St. Martin's Engineering College, Dhulapally, Secunderabad, for his support and guidance throughout our project as well as Project Coordinator, **Ms. S. SUNANDHA** Assistant Professor, Artificial Intelligence and Data Science(AI & DS) department for his valuable support.

We would like to express our sincere gratitude and indebtedness to our project supervisor, **MR. V. VISHNU VARDHAN**, Associate Professor, Artificial Intelligence and Data Science (AI & DS), St. Martins Engineering College, Dhulapally, for his support and guidance throughout our project.

Finally, we express thanks to all those who have helped us successfully completing this project. Furthermore, we would like to thank our family and friends for their moral support and encouragement. We express thanks to all those who have helped us in successfully completing the project.

M. RAJESH (23K81A7296)

K. SREE AKHIL (23K81A7292)

CH. RAM MAHIDEEP VARMA (23K81A7279)

ABSTRACT

The demand for loans has increased significantly for both personal and business purposes. Due to limited assets, banks cannot grant loans to every applicant. And the Loan approval processes in banks often depend on manual process, which leads to delays, biases, and high default risks. So, In this project we introduces a machine learning - based system to predict loan eligibility accurately and efficiently and within less time. Using a datasets including features like income, credit history, employment status, loan amount, and tenure. In this we applied supervised learning algorithms including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines. Data preprocessing involves in handling missing values, encoding categorical variables.

This project presents a Loan Eligibility Prediction System using Machine Learning, which aims to predict whether a loan applicant is eligible for a loan based on their financial and personal information. The system analyses historical loan data and uses machine learning algorithms to identify patterns that influence loan approval decisions. The dataset used in this project includes various attributes such as applicant income, co-applicant income, loan amount, loan term, credit history, education level, marital status, and property area.

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LIST OF ACRONYMS AND DEFINITIONS

S. NO	ACRONYM	DEFINITION
01	LR	Logistic Regression
02	DT	Decision Trees
03	RT	Random Forest
04	SVM	Support Vector Machines
05	XG Boost	eXtream Gradient Boosting
06	KNN	K-Nearest Neighbours
07	CSV	Comma-Separated Values
08	SQL	Structured Query Language
09	VS Code	Visual Studio Code

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CHAPTER – 1

INTRODUCTION

The Banks receive thousands of loan applications every day and making accurate and timely decisions is crucial to ensure financial stability. Traditionally, loan eligibility has been assessed manually based on income, employment status, credit history, and other financial factors. However, manual screening is time-consuming, prone to human error, and often inconsistent. With the rapid growth of data in the financial sector, machine learning has emerged as an effective solution to improve decision-making in loan approval processes.

The aim of this project is to build a predictive model that classifies loan applicants into **“Eligible” or “Not Eligible”** categories based on their demographic and financial attributes.

With the rapid advancement of data analytics and computational technologies, machine learning (ML) has emerged as a powerful tool for improving decision-making processes in the banking sector. Machine learning algorithms can analyze historical loan data, identify hidden patterns, and generate predictive models that assist banks in determining whether a loan applicant is likely to repay the loan. By leveraging features such as applicant income, credit history, employment status, loan amount, and other financial indicators, these models can provide accurate predictions regarding loan eligibility.

1.1 OBJECTIVE OF THE PROJECT:

- To develop a machine learning-based system that predicts whether a loan applicant is eligible for a loan based on their personal and financial information.
- To analyze historical loan application data to identify patterns and factors that influence loan approval decisions.
- To implement and compare different machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine to determine the most accurate prediction model.
- To improve the efficiency and accuracy of the loan approval process by reducing manual intervention and minimizing human bias.
- To assist banks and financial institutions in risk management by identifying potential loan defaults before approving loans.
- To create a reliable decision-support system that helps loan officers make faster and more consistent lending decisions.

1.2 OVERVIEW OF THE PROJECT:

The Loan Eligibility Prediction for Banks Using Machine Learning project focuses on developing an intelligent system that helps financial institutions determine whether a loan applicant is eligible for loan approval. In traditional banking systems, loan approval is often based on manual verification and predefined rules, which can be time-consuming and may lead to inconsistencies or human bias. With the increasing number of loan applications and the growing importance of risk management, banks require efficient and reliable automated systems to support their decision-making process.

This project uses machine learning techniques to analyze historical loan data and build predictive models capable of estimating the eligibility of new applicants. The system considers several important features such as applicant income, employment status, credit history, loan amount, loan term, marital status, education level, and other relevant financial indicators. By training machine learning algorithms on past data, the system learns patterns associated with approved and rejected loan applications.

The project involves several stages, including data collection, data preprocessing, feature selection, model training, and model evaluation. During preprocessing, missing values are handled and categorical variables are converted into numerical formats suitable for machine learning models. Multiple algorithms such as Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines can be applied to determine which model provides the best prediction accuracy.

Once the model is trained and validated, it can be used to predict the loan eligibility of new applicants by analyzing their input data. The system provides a prediction indicating whether the loan should be approved or not, thereby assisting banks in making faster and more consistent decisions.

CHAPTER – 2

LITERATURE SURVEY

Loan eligibility prediction has become an important research area in the banking and financial sector due to the rapid increase in loan applications and the need to minimize financial risk. Many researchers have explored the use of machine learning algorithms to automate the loan approval process and improve decision-making accuracy.

Several studies have focused on comparing different machine learning techniques for predicting loan approval. A study by Krishna raj et al. (2024) analysed models such as Logistic Regression, Decision Tree, and Random Forest for loan approval prediction. The research highlighted that machine learning can significantly improve efficiency and fairness in evaluating loan applicants by analysing historical data and identifying patterns in borrower behaviour. The results showed that Logistic Regression achieved slightly better performance compared to other models in their experiments.

Another research work proposed the use of the Random Forest algorithm to determine loan eligibility by analysing factors such as applicant income, credit history, loan amount, and employment details. The study emphasized that machine learning models can assist banks in identifying reliable borrowers and reducing the chances of loan defaults by analysing large datasets more effectively than traditional methods.

Athreya's et al. (2022) conducted a comparative study of various machine learning algorithms for predicting loan default and eligibility. Their research reviewed several approaches including k-Nearest Neighbour (KNN), Random Forest, and Support Vector Machines (SVM). The study concluded that machine learning models provide reliable predictions for credit risk analysis and can help financial institutions make safer lending decisions.

Recent systematic reviews on credit risk prediction have also highlighted the growing role of machine learning in the financial sector. These studies show that algorithms such as Logistic Regression, KNN, Decision Trees, Random Forest, and Gradient Boosting are widely used to evaluate borrower risk and predict loan approval. Evaluation metrics like accuracy, precision, recall, and F1-score are commonly used to measure model performance. However, challenges such as data imbalance, feature selection, and model interpretability still remain important research issues.

Overall, previous studies demonstrate that machine learning techniques can significantly enhance the loan approval process by automating credit assessment, reducing manual effort, and improving prediction accuracy. These findings motivate the development of intelligent loan eligibility prediction systems that support banks in making faster, data-driven lending decisions.

CHAPTER - 3

SYSTEM ANALYSIS AND DESIGN

EXISTING SYSTEM:

In the traditional banking system, loan approval is generally carried out manually by loan officers. The decision is made based on predefined rules and the evaluation of various documents submitted by the applicant, such as income proof, employment details, credit history, and other financial records.

DISADVANTAGES/LIMITATIONS OF THE EXISTING SYSTEM:

- The process is time-consuming due to manual verification.
- Decisions may be influenced by human bias or errors.
- Handling a large number of loan applications becomes inefficient.
- There is a higher risk of incorrect loan approval or rejection.
- The system lacks the ability to analyse large historical datasets effectively.

PROPOSED SYSTEM:

The proposed system uses machine learning algorithms to automate the loan eligibility prediction process. It analyses historical loan data to learn patterns and relationships between applicant attributes and loan approval outcomes. Based on this trained model, the system can predict whether a new applicant is eligible for a loan.

The system considers several input parameters such as applicant income, credit history, employment status, loan amount, loan term, marital status, education level, and property area. These features are used by the machine learning model to generate predictions.

ADVANTAGES OF THE PROPOSED SYSTEM:

- Faster processing of loan applications.
- Improved accuracy in predicting loan eligibility.
- Reduction in human intervention and bias.
- Better risk management for banks.
- Ability to handle large datasets efficiently.

CHAPTER - 4
SYSTEM REQUIREMENTS & SPECIFICATIONS

4.1 DATABASE:

The database used in the Loan Eligibility Prediction System contains historical information about loan applicants and their loan approval status. This data is used to train and evaluate machine learning models that predict whether a new applicant is eligible for a loan.

The dataset includes both financial and personal attributes of applicants, which help the model understand patterns and relationships that influence loan approval decisions. The dataset consists of multiple records, where each record represents a loan application submitted by a customer. Each record contains several attributes such as applicant income, loan amount, credit history, and property area. These attributes serve as input features for the machine learning models.

The final attribute in the dataset is the Loan Status, which indicates whether the loan application was approved or rejected. This attribute acts as the target variable for prediction.

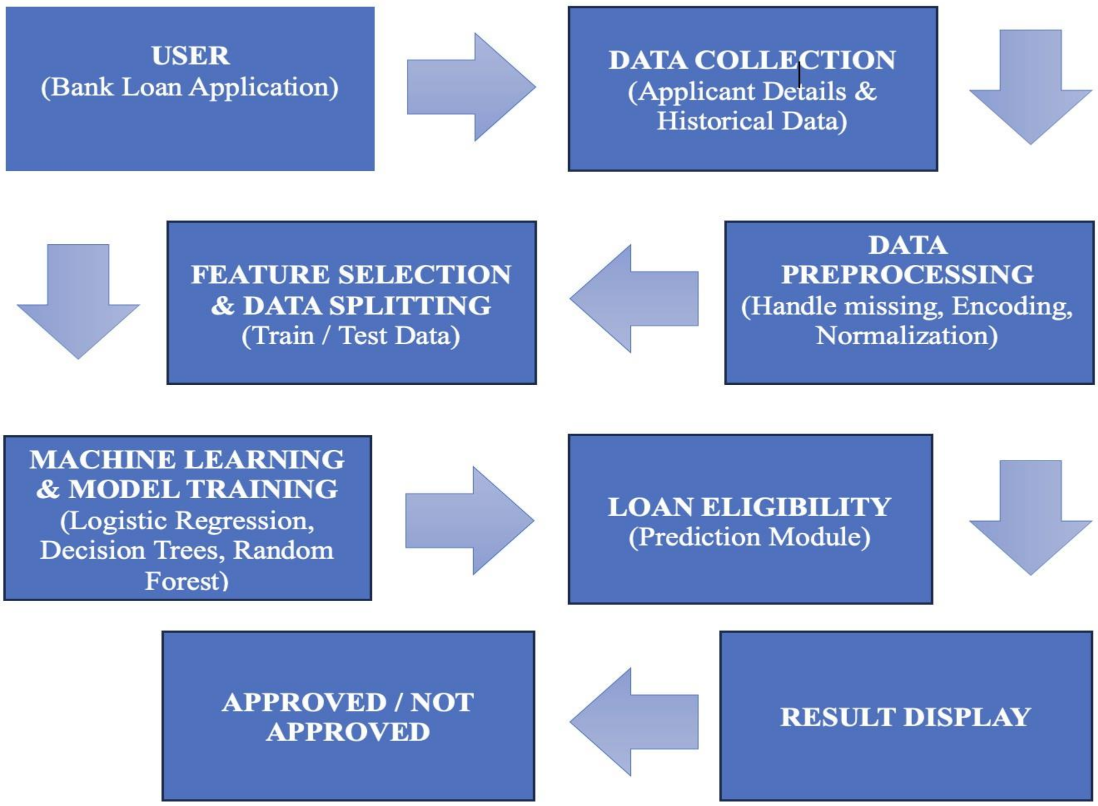
Table Name: `loan_data`

COLUMN NAME	DATA TYPE	DESCRIPTION
Loan_ID	String	Unique ID for each loan application
Gender	String	Gender of the applicant (Male/Female)
Married	String	Marital status of the applicant
Dependents	Integer	Number of dependents
Education	String	Education level (Graduate/Not Graduate)
Self-employed	String	Whether the applicant is self-employed

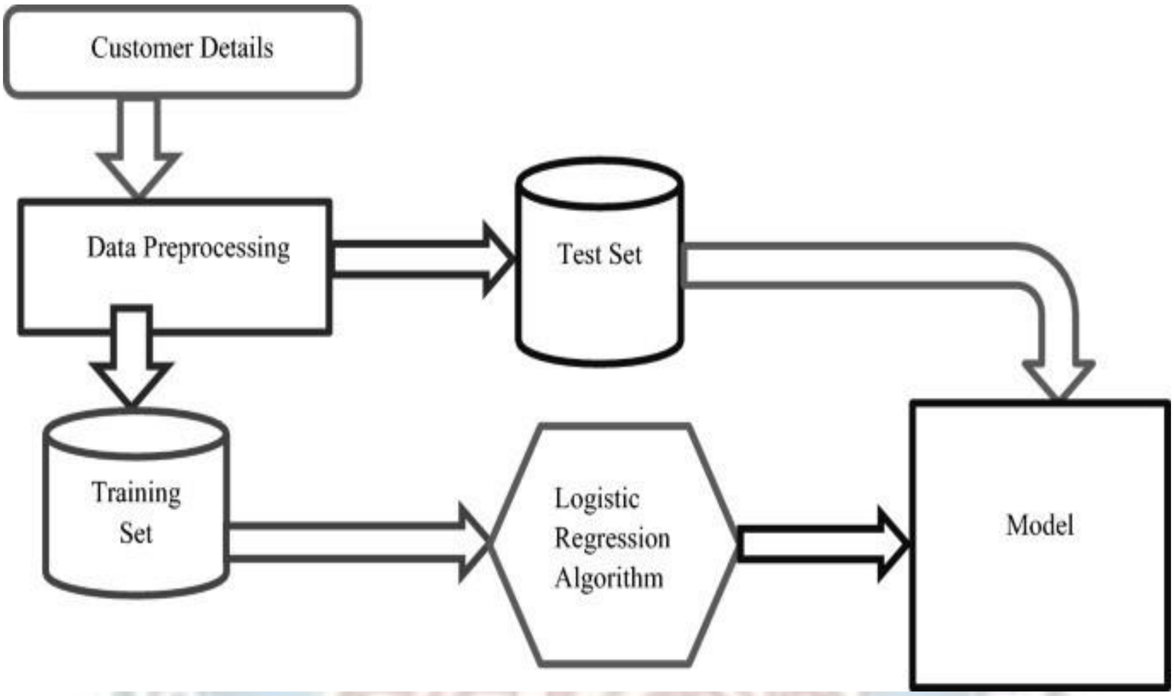
Applicant Income	Integer	Monthly income of the applicant
Compliant Income	Integer	Monthly income of co-applicant
Loan Amount	Integer	Loan amount requested
Loan_Amount_Term	Integer	Loan repayment term (in months)
Credit_History	Integer	Credit history (1 = good, 0 = bad)
Property_Area	String	Area type (Urban/Semiurban/Rural)
Loan_Status	String	Loan approved or not (Y/N)

4.2 DESIGN:

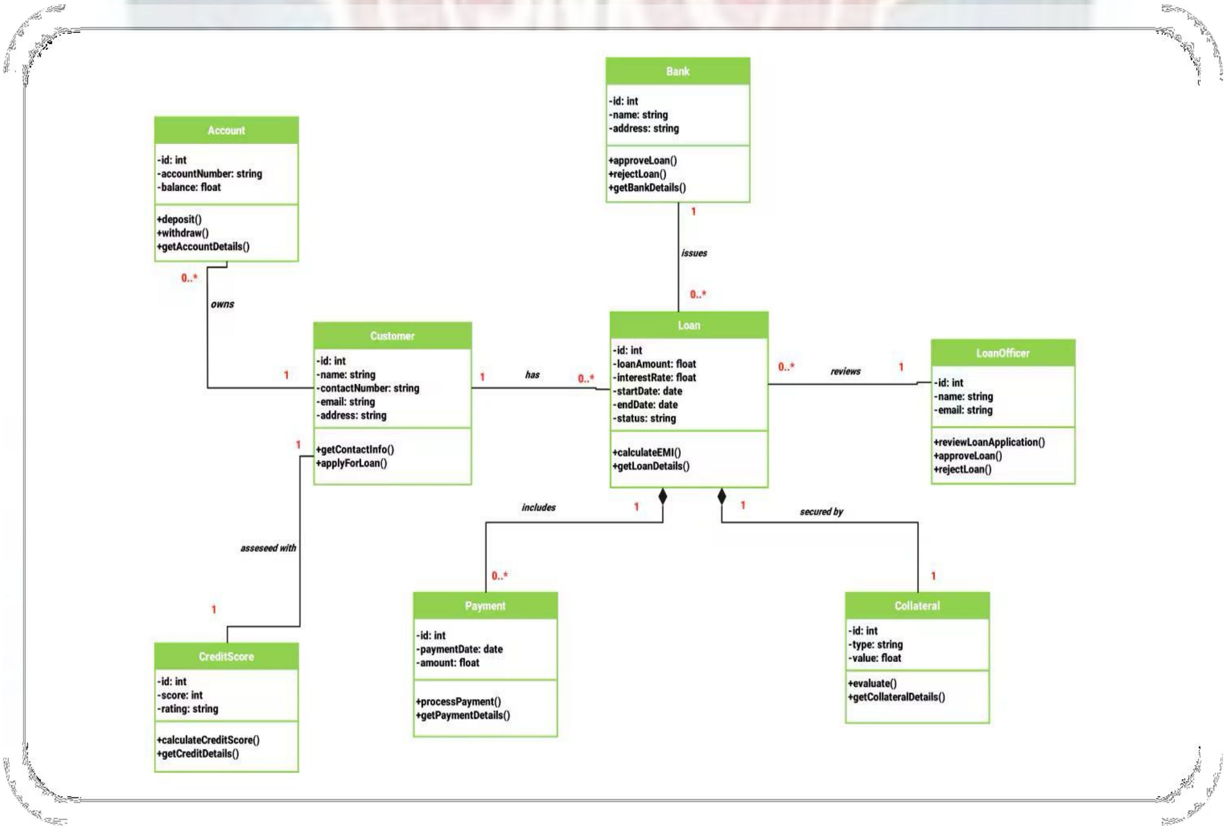
4.2.1 SYSTEM ARCHITECTURE:



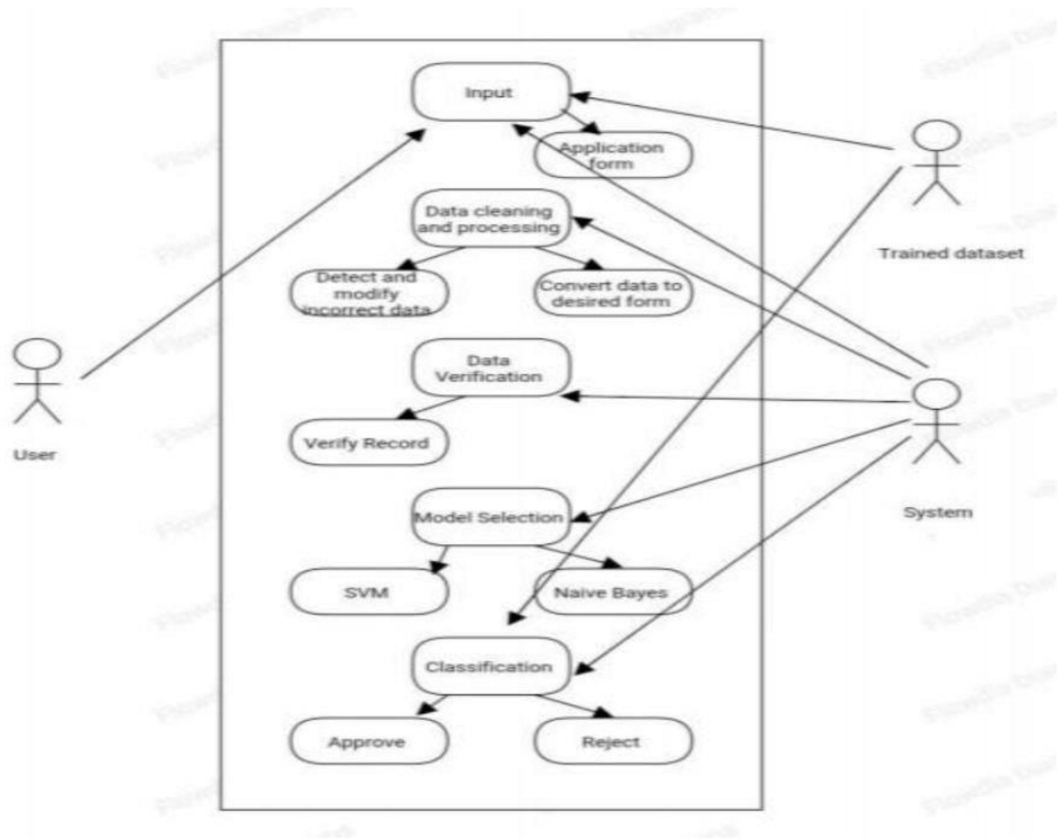
4.2.2 DATA FLOW DIAGRAM:



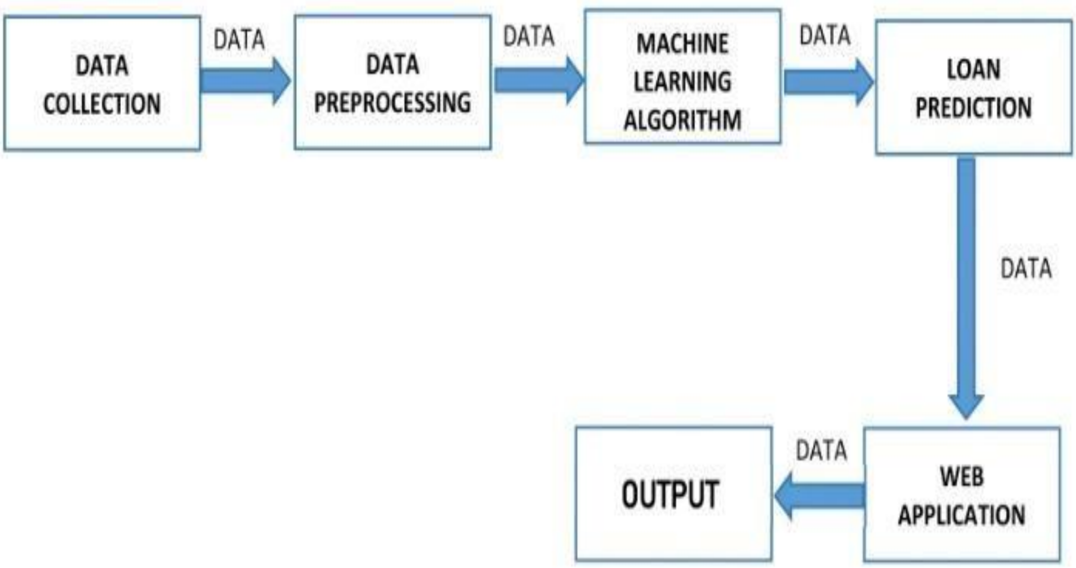
4.2.3 CLASS DIAGRAM:



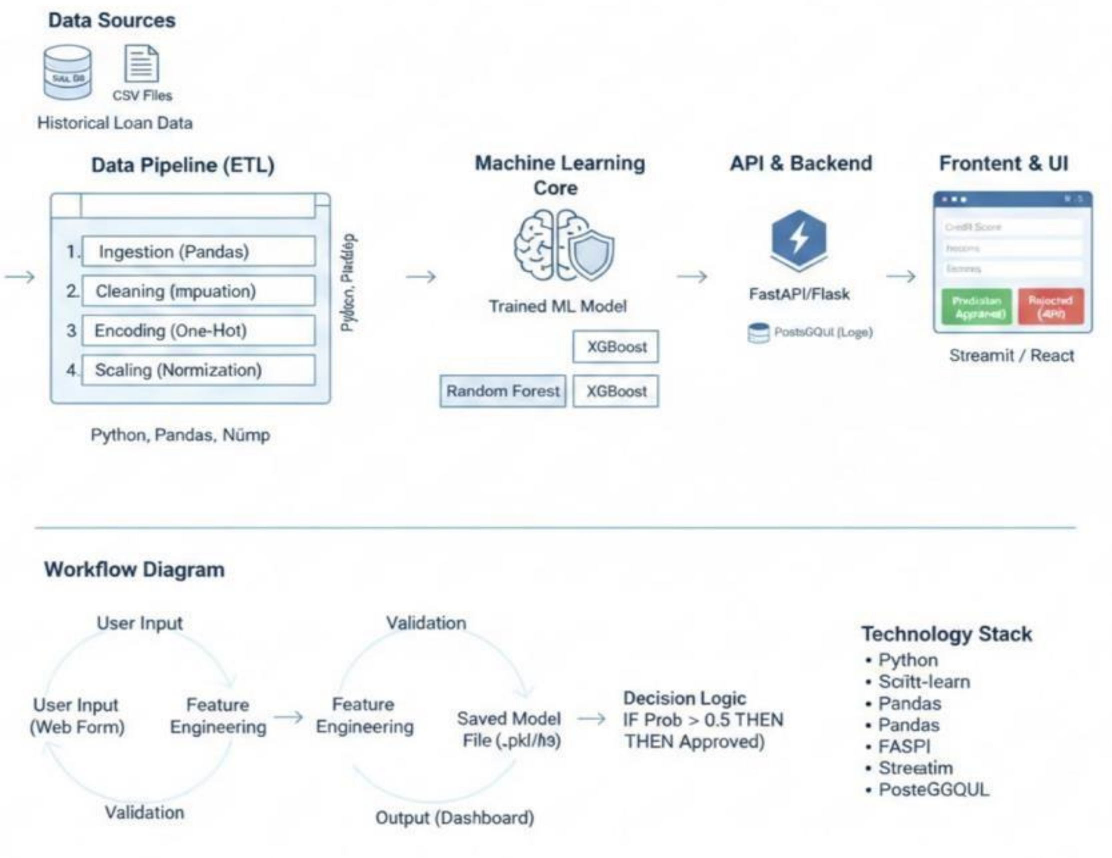
4.2.4 USE CASE DIAGRAM:



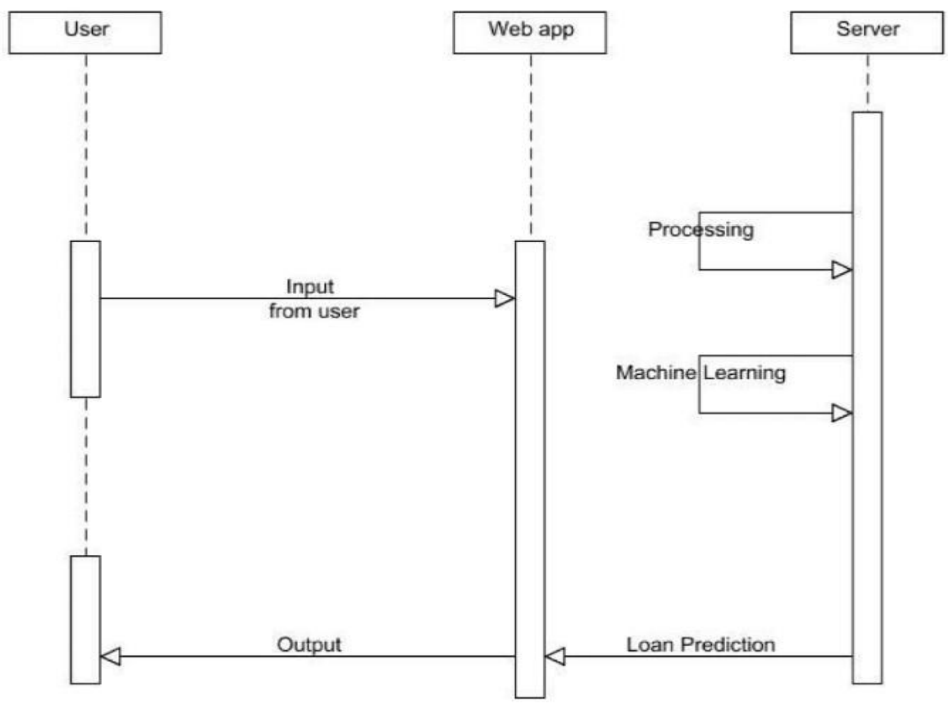
4.2.5 COLLABORATION DIAGRAM:



4.2.6 COMPONENT DIAGRAM:



4.2.6 DEPLOYMENT DIAGRAM:



4.3 SYSTEM REQUIREMENTS:

The Loan Eligibility Prediction System using Machine Learning requires both hardware and software resources to develop, train, and run the predictive model effectively. These requirements ensure that the system operates efficiently and provides accurate results.

4.3.1 HARDWARE REQUIREMENTS:

The following hardware components are required for the implementation of the system:

- **Processor:** Intel Core i3 or higher.
- **RAM:** Minimum 4 GB (8 GB recommended for better performance).
- **Storage:** At least 20 GB of free disk space.
- **System Type:** 64-bit computer system.
- **Input Devices:** Keyboard and mouse.

4.3.2 SOFTWARE REQUIREMENTS:

The following software tools and technologies are required to develop and run the project:

- **Operating System:** Windows, Linux, or macOS.
- **Programming Language:** Python.
- **Development Environment:** Jupiter Notebook, Anaconda, or VS Code.

LIBRARIES/FRAMEWORKS:

- **NumPy** (for numerical computations).
- **Pandas** (for data manipulation and analysis).
- **Scikit-learn** (for implementing machine learning algorithms).
- **Matplotlib / Seaborn** (for data visualization).

***DATABASE:** CSV files or SQLite for storing dataset information.

4.4 TESTING:

Testing is an important phase in software development that ensures the system works correctly and meets the required objectives. In this project, different types of testing were performed to verify that the loan eligibility prediction system functions accurately and efficiently. The testing process helps identify errors, improve system reliability, and ensure that the prediction results are correct.

4.4.1 SYSTEM TESTING:

System testing is performed to evaluate the complete and integrated system. It verifies whether the entire loan eligibility prediction system works properly according to the project requirements. In this project, system testing was conducted by providing different applicant data as input and checking whether the system correctly predicts the loan eligibility status. The system was tested for various cases including valid inputs, missing data, and edge conditions.

The results showed that the system successfully processes applicant information, applies machine learning models, and generates accurate loan approval predictions. This testing ensures that the overall system performs efficiently and produces reliable results.

4.4.2 MODULE TESTING:

Module testing focuses on testing individual components or modules of the system independently. Each module is tested separately to ensure that it performs its intended function without errors.

In the loan eligibility prediction system, several modules were tested individually, including:

- Data Collection Module
- Data Preprocessing Module
- Feature Selection Module
- Model Training Module
- Prediction Module

Each module was verified to ensure that it processes the input data correctly and produces the expected output before integrating it with other modules.

4.4.3 INTEGRATION TESTING:

Integration testing is performed after module testing to ensure that all modules work together correctly. It checks the interaction and data flow between different modules of the system. In this project, integration testing was conducted by combining the data preprocessing module, machine learning model module, and prediction module. The data was passed through each stage to verify that the output of one module correctly becomes the input for the next module. The testing confirmed that the modules are properly integrated and that the system processes the data smoothly from input to final loan prediction.

4.4.4 ACCEPTANCE TESTING:

Acceptance testing is carried out to verify whether the system meets the requirements and expectations of the end users. This testing ensures that the system performs the intended task effectively in real-world conditions. For this project, acceptance testing involved evaluating whether the loan eligibility prediction system provides accurate and meaningful results for different loan applications. The system was tested with multiple sample inputs to ensure that it predicts loan approval or rejection correctly. The successful results indicate that the system satisfies the project objectives and can be used as a supportive tool for banks to assist in loan approval decision-making.

CHAPTER - 5

SOURCE CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


loan = pd.read_csv
('/content/drive/MyDrive/dataset/loan_approval_dataset.csv')
loan.sample(5)


from google.colab import drive
drive.mount('/content/drive')


loan.shape


loan.info()


loan.describe()


loan.rename(columns=lambda x: x.lower().strip(), inplace=True)
print(loan.columns)


# Checking the datatypes of the columns
loan.dtypes


# In the loan dataset their is no Null values
# If in case is their any null values is their for Categorial values change
with Mode/ffill/bfill and for Numerical value change with mean/median as
per the data requirements
loan.isnull().sum()


loan['loan_status'].value_counts().plot.bar()


loan['no_of_dependents'].value_counts(normalize=True).plot.pie(title=
'no_of_dependents',autopct='%1.1f%%',figsize=(8, 5))
```

```
education=pd.crosstab(loan['education'],loan['loan_status'])
education
```

```
education.plot(kind="bar", stacked=True, figsize=(6,6))
plt.show()
```

```
self_employed=pd.crosstab(loan['self_employed'],loan['loan_status'])
self_employed
```

```
self_employed.plot(kind="bar", stacked=True, figsize=(6,6))
plt.show()
```

```
Dependents=pd.crosstab(loan['no_of_dependents'],loan['loan_status'])
Dependents
```

```
Dependents.plot(kind="bar", stacked=True, figsize=(6,6))
plt.show()
```

```
loan['cibil_score'].hist(bins=50)
```

```
numerical_cols = loan.select_dtypes(include=['int64', 'float64']).columns
plt.figure(figsize=(15, 5))
loan[numerical_cols].boxplot(rot=45, grid=True)
```

```
# Checking the outliers for the numerical columns
numerical_cols = loan.select_dtypes(include=['int64', 'float64']).columns
fig, axes = plt.subplots(2, 3, figsize=(16, 8), constrained_layout=True)  #
2 rows, 3 columns
for ax, column in zip(axes.flat, numerical_cols):
    sns.histplot(data=loan, x=column, ax=ax, kde=True, color='gray',
bins=15)  # Add KDE if desired
    ax.set_title(f'Histogram for {column}', fontsize=10)
    ax.set_xlabel('')
```

Label Encoding Processing

```
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

Convert Categorical Variables to Numerical

```
lc = LabelEncoder()
for col in ['education','self_employed','loan_status']:
    loan[col] = lc.fit_transform(loan[col])
```

Standarizing the Numerical Columns

```
scaler = StandardScaler()
for col in
['no_of_dependents','income_annum','loan_amount','loan_term','cibil_score',
'residential_assets_value','commercial_assets_value','luxury_assets_value',
'bank_asset_value']:
   loan[[col]] = scaler.fit_transform(loan[[col]])
```

```
# Stastical Summary
loan.describe()
```

Train-Test Spliting the Data

```
x = loan.iloc[:,1:12]
x.shape
```

```
(4269, 11)
```

```
y = loan[['loan_status']]
y.shape
```

```
(4269, 1)
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test =
train_test_split(x,y,test_size=0.25,random_state = 42)
```

Model Selection & Implementation

```
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
```

Logistic Regression Model

```
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression()
lr.fit(x_train, y_train)
```

```
y_pred = lr.predict(x_test)
Accuraacy_score = accuracy_score(y_test, y_pred)
Train = lr.score(x_train,y_train)*100
cm = confusion_matrix(y_test, y_pred)
print('confusion_matrix:\n',cm)
report = classification_report(y_test, y_pred)
print('classification_report:\n',report)
print('Training_accuracy: ', Train)
print('Testing_accuracy: ',Accuraacy_score)
```

```
from sklearn.tree import DecisionTreeClassifier
dt = DecisionTreeClassifier()
dt.fit(x_train, y_train)
```

DecisionTreeClassifier()

```
y_pred = dt.predict(x_test)
Accuraacy_score = accuracy_score(y_test, y_pred)
Train = dt.score(x_train,y_train)*100
cm = confusion_matrix(y_test, y_pred)
print('confusion_matrix:\n',cm)
report = classification_report(y_test, y_pred)
print('classification_report:\n',report)
print('Training_accuracy: ', Train)
print('Testing_accuracy: ',Accuraacy_score)
```

KNN Model

```
from sklearn.neighbors import KNeighborsClassifier
knc = KNeighborsClassifier()
knc.fit(x_train, y_train)
```

```
y_pred = knc.predict(x_test)
Accuraacy_score = accuracy_score(y_test, y_pred)
Train = knc.score(x_train,y_train)*100
cm = confusion_matrix(y_test, y_pred)
print('confusion_matrix:\n',cm)
report = classification_report(y_test, y_pred)
print('classification_report:\n',report)
print('Training_accuracy: ', Train)
print('Testing_accuracy: ',Accuraacy_score)
```

```
from sklearn.naive_bayes import GaussianNB
gnb = GaussianNB()
gnb.fit(x_train, y_train)
```

```
y_pred = gnb.predict(x_test)
Accuraacy_score = accuracy_score(y_test, y_pred)
Train = gnb.score(x_train,y_train)*100
cm = confusion_matrix(y_test, y_pred)
print('confusion_matrix:\n',cm)
report = classification_report(y_test, y_pred)
print('classification_report:\n',report)
print('Training_accuracy: ', Train)
print('Testing_accuracy: ',Accuraacy_score)
```

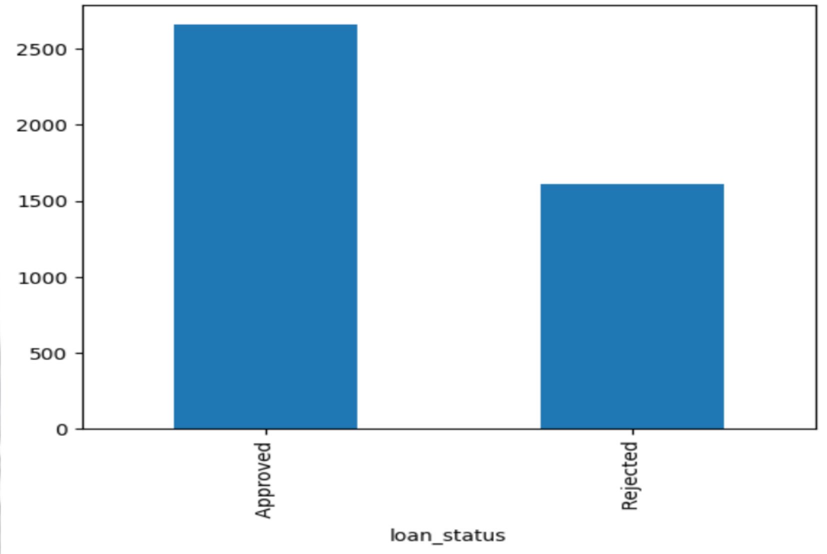
```
from sklearn.svm import SVC
svc = SVC()
svc.fit(x_train, y_train)
```

```
y_pred = svc.predict(x_test)
Accuraacy_score = accuracy_score(y_test, y_pred)
Train = svc.score(x_train,y_train)*100
cm = confusion_matrix(y_test, y_pred)
```

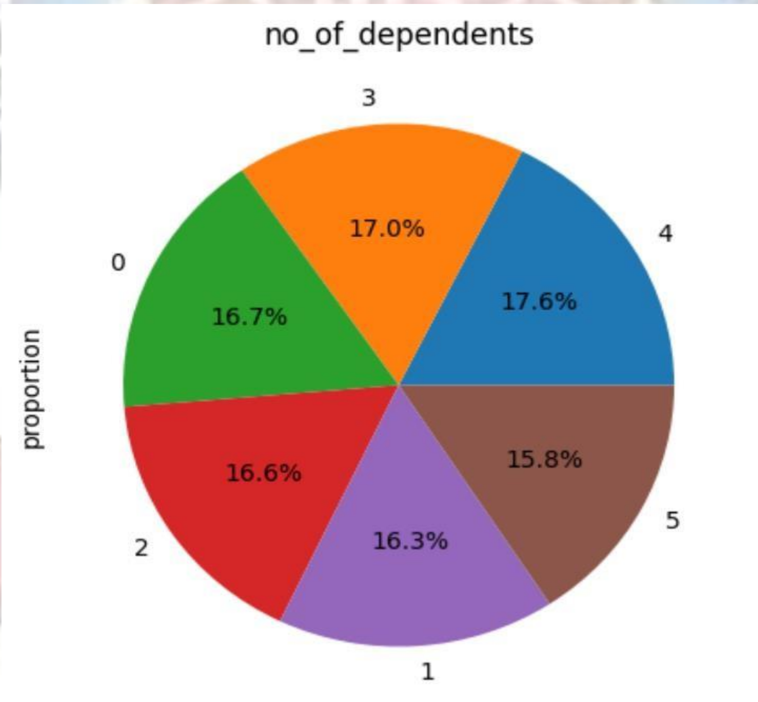
```
print('confusion_matrix:\n',cm)
report = classification_report(y_test, y_pred)
print('classification_report:\n',report)
print('Training_accuracy: ', Train)
print('Testing_accuracy: ',Accuraacy_score)


models = [ 'Logistic Regression','Decision Tree','K-Nearest
Neighbors','Naive Bayes','SVM']
train_accuracy = [92, 100, 95, 93, 95]
test_accuracy = [90, 98, 89, 93, 93]
x = np.arange(len(models))
width = 0.35
fig, ax = plt.subplots(figsize=(15,8))

# Plot bars
bars1 = ax.bar(x - width/2, train_accuracy, width, label='Training
Accuracy', color='tab:blue')
bars2 = ax.bar(x + width/2, test_accuracy, width, label='Testing Accuracy',
color='tab:orange')
ax.set_xlabel('Models')
ax.set_ylabel('Accuracy')
ax.set_title('Model Accuracy Comparison')
ax.set_xticks(x)
ax.set_xticklabels(models, rotation=30, ha="right")
ax.legend()
plt.show()
```

CHAPTER - 6

EXPERIMENTAL RESULTS

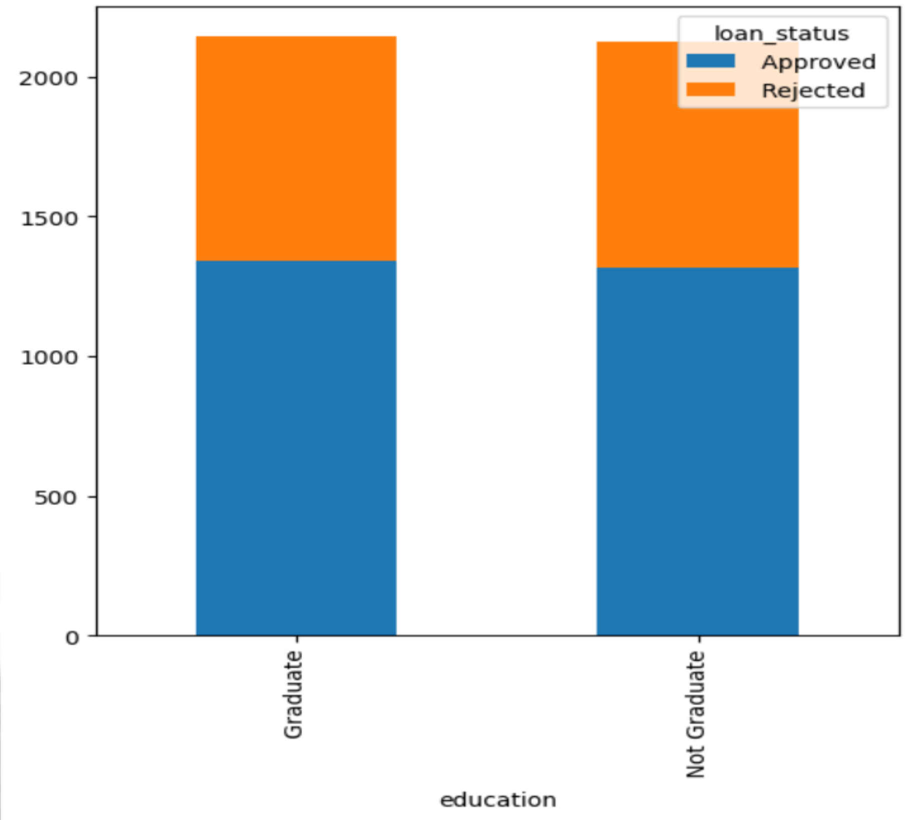


LOAN STATUS (Bar Graph)



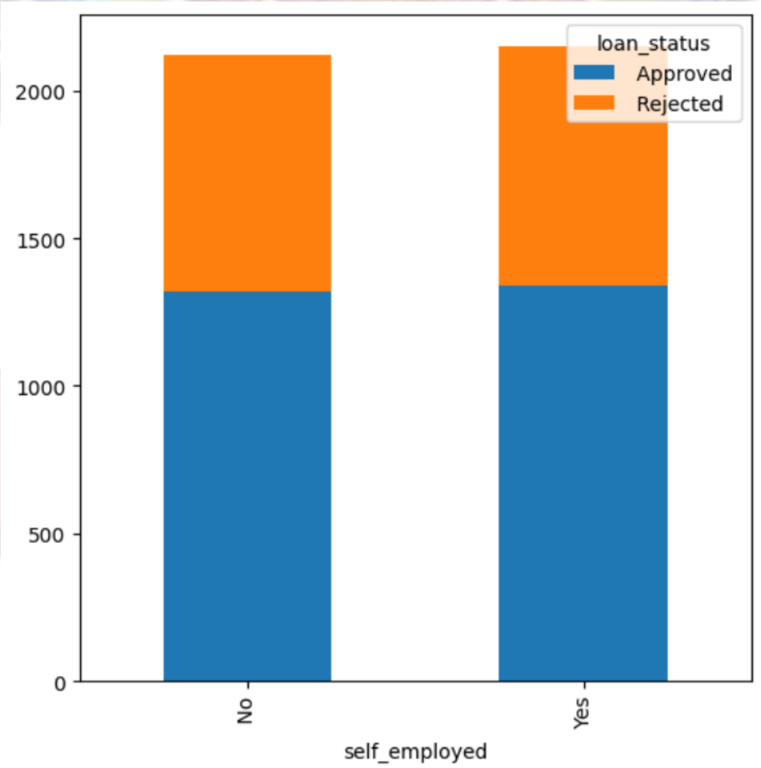
COMPARISON OF DEPENDENTS

(Pie Chart)



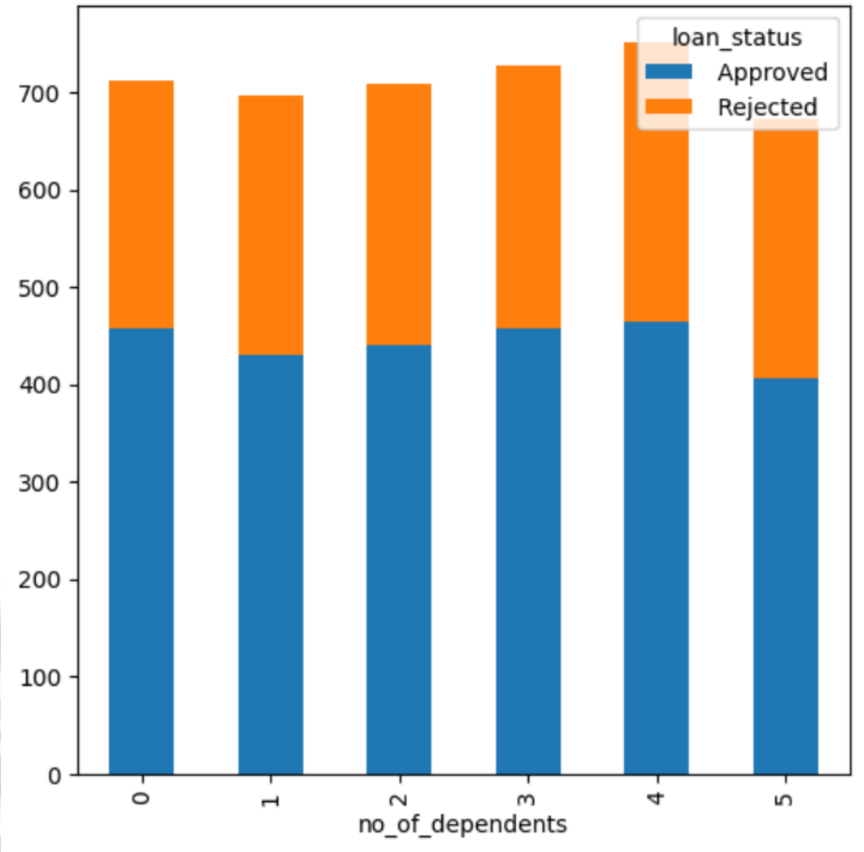
COMPARISION IN EDUCATION

(Stacked Bar Chart)



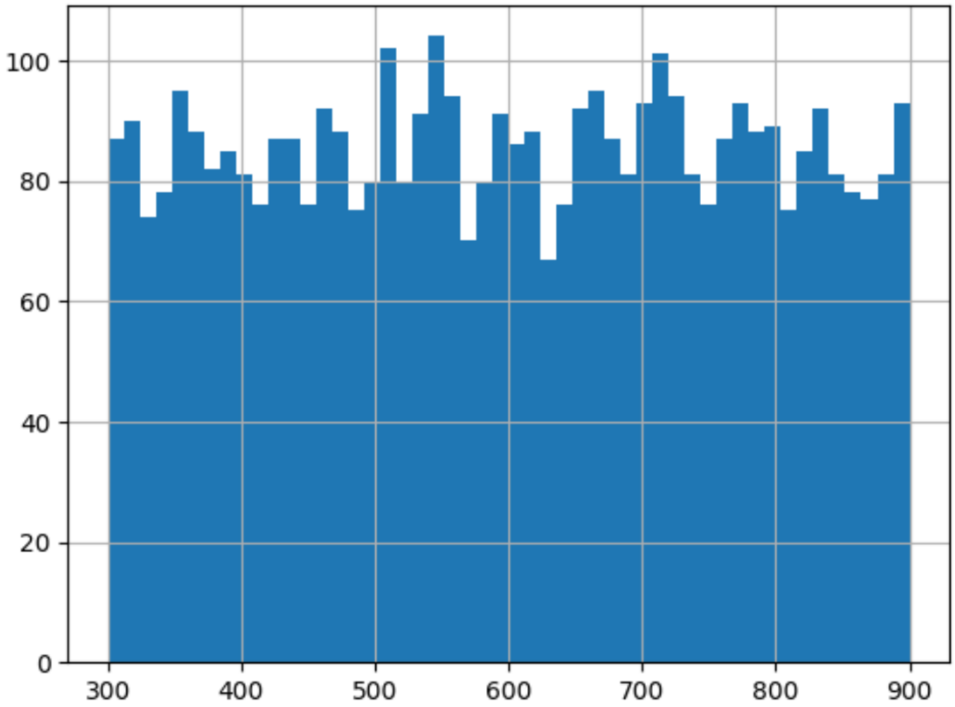
COMPARISION IN SELF EMPLOYED

(Stacked Bar Chart)



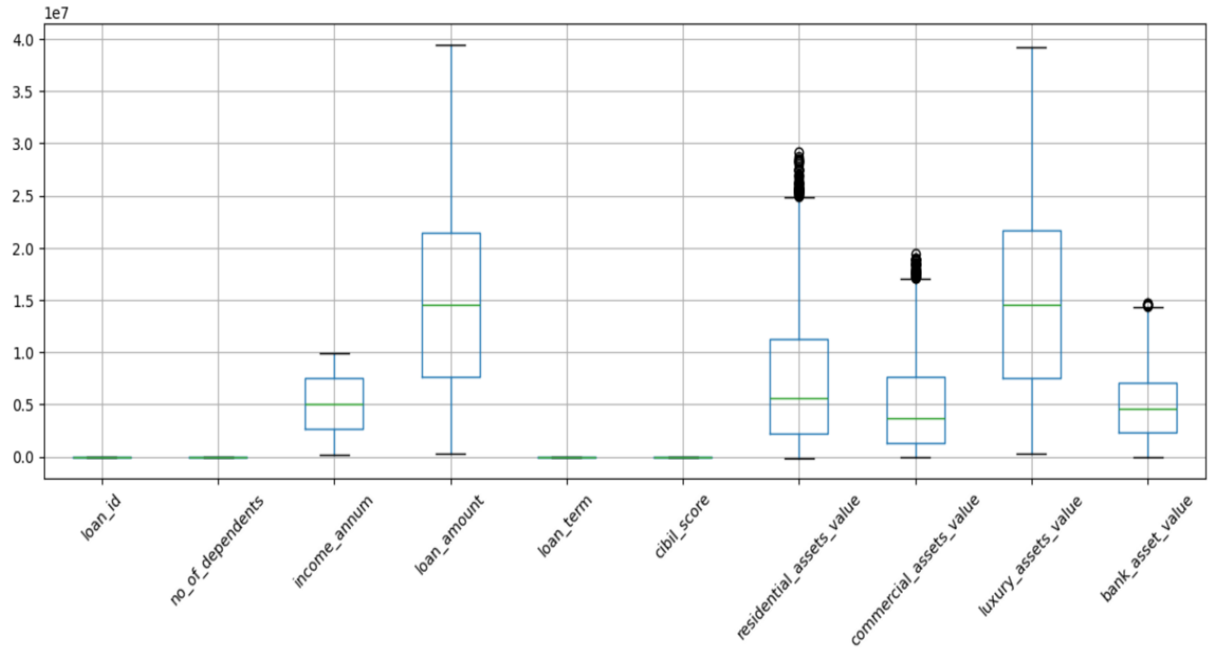
NO. OF DEPENDENTS

(Stacked Bar Chart)



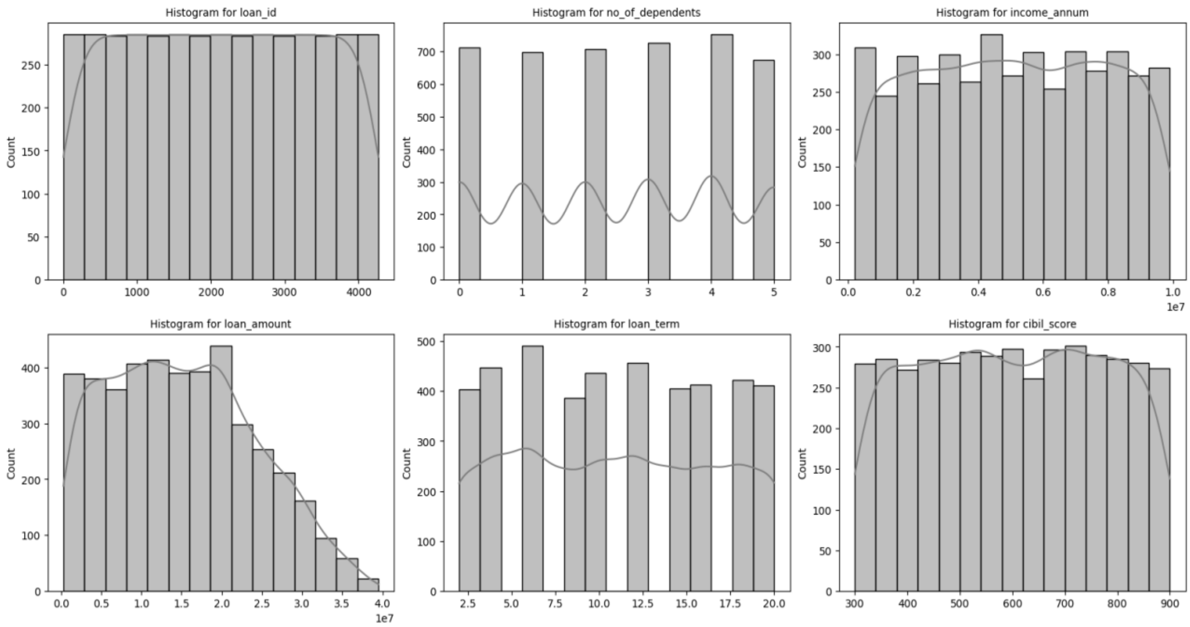
DISTRIBUTION OF THE VARIABLES

(Histogram)



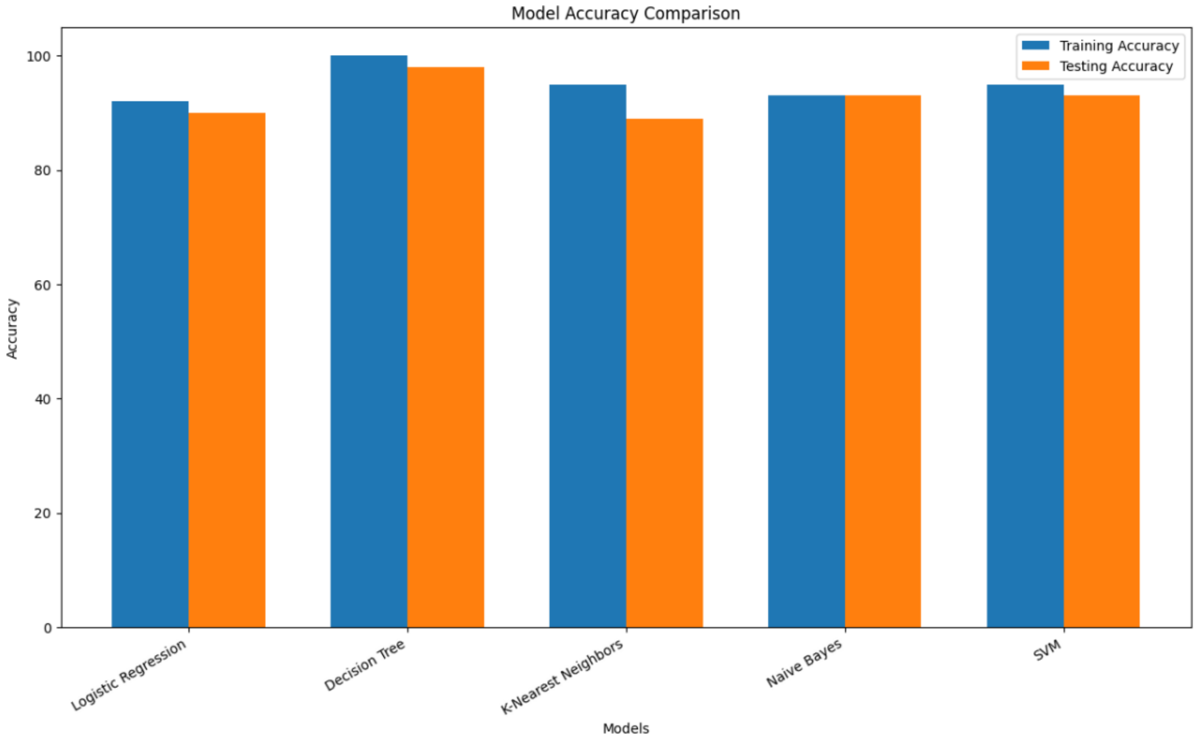
CHECKING OUTLIERS FOR NUMERICAL COLUMNS

(Box Plot)



REPRESENTATION OF ALL DATA SETS

(Histogram)



COMPARISON OF ALL MODELS

(Grouped Bar Chart)

CHAPTER - 7

CONCLUSION & FUTURE ENHANCEMENTS

7.1 CONCLUSION:

The Loan Eligibility Prediction System using Machine Learning was developed to assist banks and financial institutions in making accurate and efficient loan approval decisions. Traditional loan approval processes are often time-consuming and rely heavily on manual evaluation, which can lead to delays and human errors. By integrating machine learning techniques, the proposed system provides a more reliable and automated approach for evaluating loan applications. In this project, historical loan data is analysed and processed using machine learning algorithms to identify patterns related to loan approval. Various applicant attributes such as income, credit history, employment status, and loan amount are used as input features to train the predictive model. The trained model can then predict whether a new applicant is eligible for a loan or not.

The system improves the efficiency of the loan approval process by reducing manual effort, minimizing the chances of errors, and enabling faster decision-making. It also helps banks manage financial risk by identifying potentially risky applicants before approving loans. Overall, the implementation of machine learning in loan eligibility prediction demonstrates the potential of data-driven approaches in improving financial decision-making.

7.2 FUTURE ENHANCEMENTS:

Although the proposed system performs effectively, several improvements can be implemented in the future to enhance its capabilities:

- **Integration with Real-Time Banking Systems:** The model can be integrated with existing banking databases to automatically fetch applicant data and provide real-time loan approval predictions.
- **Use of Advanced Machine Learning Models:** More advanced algorithms such as Gradient Boosting, XGBoost, or Deep Learning techniques can be applied to improve prediction accuracy.

- **Incorporation of Credit Score APIs:** External credit score systems can be integrated to provide more reliable financial data for decision-making.
- **Development of a Web-Based or Mobile Application:** The system can be converted into a user-friendly web or mobile application for easier access by bank employees and customers.
- **Improved Data Security and Privacy:** Future versions can include stronger security mechanisms to protect sensitive financial and personal data.
- **Handling Larger and More Diverse Datasets:** The model can be trained on larger datasets from multiple financial institutions to improve its generalization capability and robustness.

With these enhancements, the system can become a more powerful and comprehensive tool for financial institutions, enabling smarter and more secure loan approval processes.

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